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Optimising EcoPack Ltd's Logistics Model

ECO PACK LTD

Introduction

EcoPack Ltd is a UK compostable packaging supplier that operates a zero-landfill policy. Signing of a national distribution contract with two major retailers has necessitated redesign of its logistics network. Internal figures indicate a 12% decrease in sustainability KPIs and an 18% increase in late deliveries. This study utilises Linear Programming and Decision Analysis to assess three logistics network options.

Model Development

Decision variables included: Total Cost, CO_2 Emissions, Delivery Delay, Customer Satisfaction, Sustainability Score and Monthly Volume.

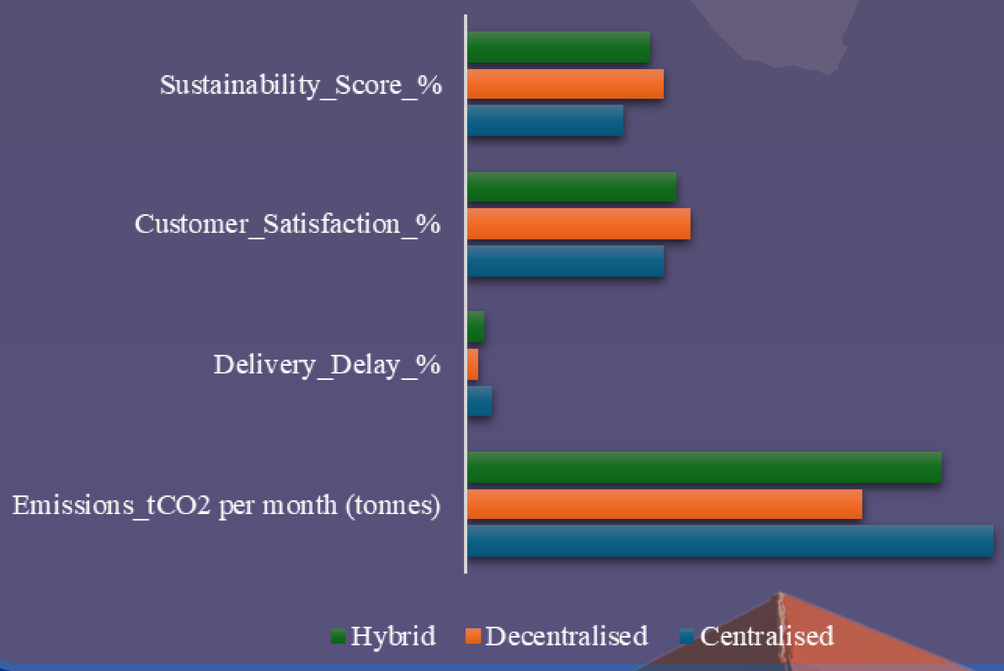
Justification of Models

- Linear Programming will pinpoint the least expensive model that clears established operational and environmental hurdles. LP works well when there is one goal you want to quantify (Hillier and Lieberman, 2015).
- Decision Analysis will score each option across all weighted criteria at the same time. Decision Analysis allows stakeholders to capture trade-offs that LP cannot account for (Triantaphyllou, 2000).

Linear Programming

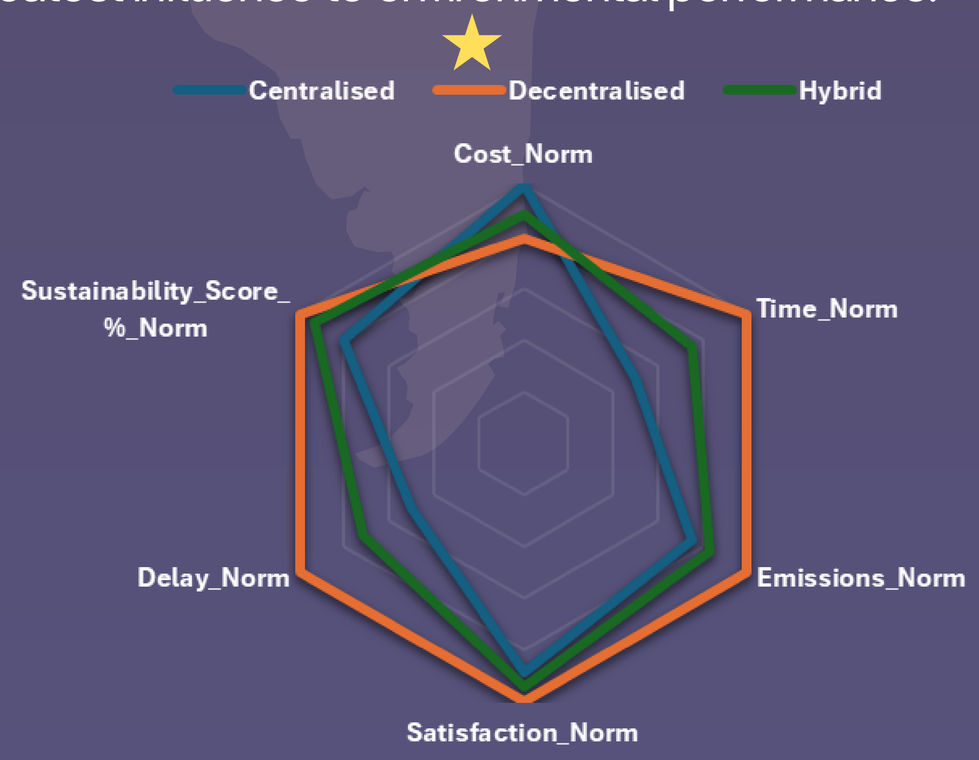
Linear Programming suggested the Hybrid model ($X_3 = 1$) as the cost-optimal feasible solution at £170,000 per month, while Centralised was disqualified on 4 constraints as depicted below; Hybrid cleared every criterion at a lower cost than Decentralised.

Hybrid	Decentralised	Centralised
£170,000	£190,000	£150,000



Decision Analysis

Decision Analysis was applied using a weighted scoring matrix to evaluate logistic model performance across all criteria simultaneously, rather than only identifying the cheapest feasible option. Criteria weights were given reflecting EcoPack's sustainability-first mission, giving greatest influence to environmental performance.



Conclusion & Recommendation

Linear Programming found Hybrid to be the feasible solution with the lowest monthly cost: £170,000. Decision Analysis found Decentralised to have the highest overall weighted score: 0.97 — ranking highest for sustainability, emissions and satisfaction.

We suggest **Decentralised** be selected. The 12% drop in sustainability for EcoPack means the additional £20,000 per month is an investment rather than a luxury.

LP gives us what is feasible, DA gives us what is right — when LP and DA conflict, mission comes into play.

References

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